



ANNUAL MONITORING REPORT

Riverbend Watershed

Water quality, flow and habitat, 2026

4 monitoring stations	11,842 samples analysed	-18% nitrate vs. 2024	94% data completeness
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Summary

Water quality in the Riverbend catchment continued to improve through 2026, extending the trend first recorded after the riparian buffer programme began in 2023. Mean nitrate concentration fell at every station, and two of the four now sit below the 5 mg/L management target for the first time since monitoring began. Dissolved oxygen remained within the healthy range for 96 percent of readings.

The exception is station RB-7, downstream of the Millford confluence, where nitrate remains elevated and summer temperature peaks are increasing year on year. Section 4 sets out the proposed response.

Report	Period	Prepared by	Status
RCA-2026-04	Jan – Dec 2026	Monitoring & Science Unit	Approved for release

In this report

- Catchment description and the four-station network.
- Flow, rainfall and the 2026 low-flow period.
- Nitrate, phosphate, oxygen and temperature results.
- Recommended actions and their estimated cost.

Publication details

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1. Catchment and monitoring network

The Riverbend catchment drains 412 square kilometres of mixed arable and pasture land before joining the Kelder estuary. Four permanent stations record stage, temperature and conductivity at fifteen-minute intervals; grab samples for laboratory analysis are collected fortnightly, and weekly during the March–May high-flow window.

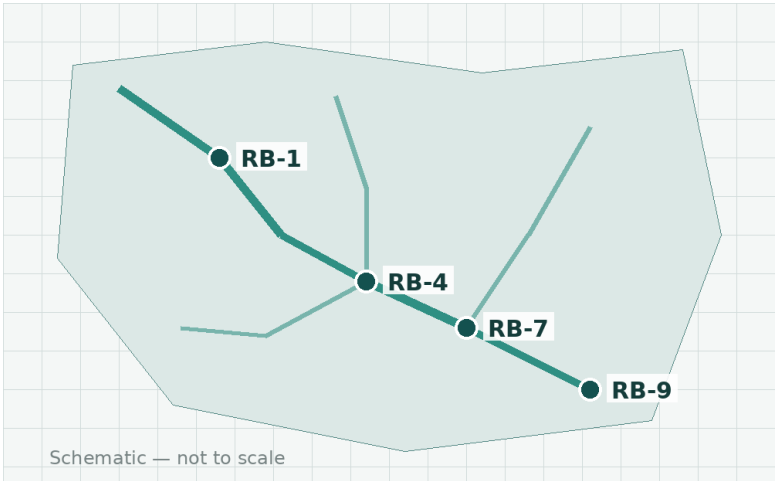


Figure 1 — Monitoring network. Stations RB-1 to RB-9 along the main channel and two tributaries.

Station summary

Station	Location	Elevation	Catchment	Record since
RB-1	Upper Riverbend, above Hollowbrook	218 m	46 km ²	2011
RB-4	Mid-catchment, Thorne Bridge	96 m	154 km ²	2011
RB-7	Below Millford confluence	41 m	311 km ²	2016
RB-9	Tidal limit, Kelder Reach	6 m	412 km ²	2019

Why RB-7 matters

RB-7 integrates every upstream input, so it is the station that determines compliance for the catchment as a whole. Improvements elsewhere are not visible in the statutory return until they appear here.

2. Flow and rainfall

Total rainfall for the year was 795 mm, close to the ten-year mean of 781 mm, but its distribution was unusually concentrated: April alone accounted for 15 percent of the annual total. Discharge tracked rainfall with a lag of two to four days at RB-4, and the late-summer low-flow period was the longest recorded, running from 22 July to 3 September.

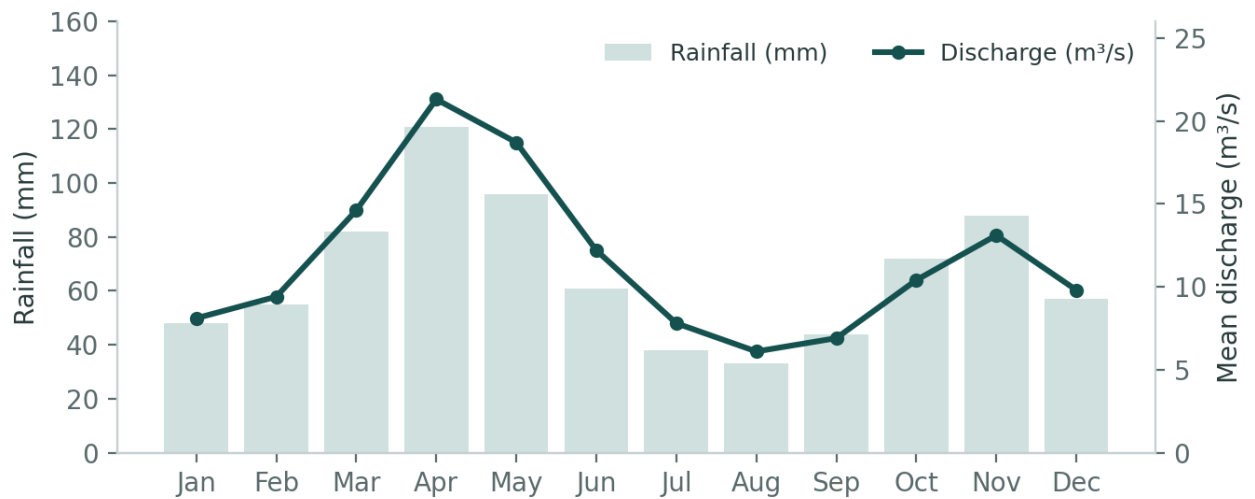


Figure 2 — Monthly rainfall and mean daily discharge at RB-4, 2026.

3. Chemistry and dissolved oxygen

Nitrate (NO_3^-) declined at all four stations for the third consecutive year. Dissolved oxygen shows the expected inverse relationship with temperature; the fitted slope of $-0.32 \text{ mg L}^{-1} \text{ per } ^\circ\text{C}$ is consistent with saturation behaviour and gives no evidence of an additional oxygen demand from organic loading.



Figure 3 — Mean nitrate by station.

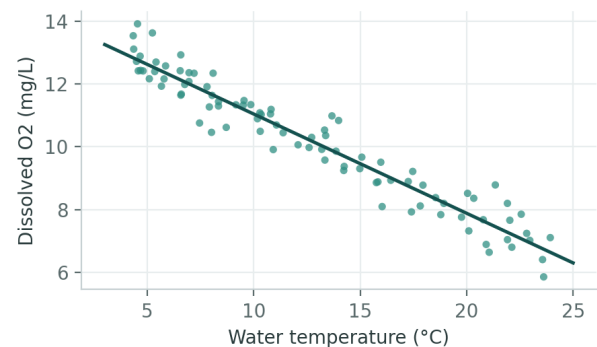


Figure 4 — Temperature vs. dissolved oxygen.

Compliance against management targets

Determinand	Target	2026 mean	Exceedances	Trend
Nitrate (mg/L)	≤ 5.0	4.9	2 of 4 stations	improving
Phosphate (mg/L)	≤ 0.10	0.08	none	improving
Dissolved O ₂ (mg/L)	≥ 7.0	9.1	18 readings	stable
Temperature (°C)	≤ 21.0	13.4	9 days at RB-7	worsening

4. Findings and recommendations

Principal findings

- Nitrate fell 18 percent against the 2024 baseline; RB-1 and RB-9 are now below target.
- Riparian buffers installed in 2023 correspond to the largest reductions, though the effect cannot be separated from three consecutive years of below-average spring rainfall.
- Summer water temperature at RB-7 exceeded 21 °C on nine days, against four in 2025 and one in 2024.
- Data completeness reached 94 percent; the shortfall is almost entirely the RB-7 sonde outage in August.

Recommended actions

1. Extend the buffer programme to the Millford tributary, targeting 6 km of bank by autumn 2027.
2. Install shade planting at the three RB-7 reaches identified in the thermal survey.
3. Replace the RB-7 multiparameter sonde and hold a calibrated spare on site.
4. Move to weekly sampling at RB-7 between June and September.

Action	Owner	Cost (est.)	Target date
Millford buffer extension	Land Management	EUR 240,000	Oct 2027
Shade planting, RB-7 reaches	Habitat Team	EUR 38,000	Mar 2027
Sonde replacement and spare	Monitoring Unit	EUR 21,500	Feb 2027
Extended summer sampling	Monitoring Unit	EUR 9,000/yr	Jun 2027

Appendix A — Analytical methods and limits

Laboratory work was carried out under ISO/IEC 17025 accreditation; limits of detection are quoted at three times the standard deviation of the blank.

Determinand	Method	Limit of detection	Uncertainty
Nitrate	Ion chromatography	0.05 mg/L	± 6%
Phosphate	Colorimetric, molybdenum blue	0.005 mg/L	± 8%
Dissolved oxygen	Optical sensor, in situ	0.1 mg/L	± 2%
Suspended solids	Gravimetric, 0.45 micron filter	1 mg/L	± 5%

Data availability and quality flags

Validated series are published quarterly. Records affected by instrument drift, fouling or the August outage are retained but flagged; flagged values are excluded from the means quoted here.

- **Flag A** — validated against a laboratory duplicate; used in all statistics.
- **Flag B** — within tolerance but un-duplicated; used in trends only.
- **Flag C** — suspect or interpolated; excluded, retained for audit.

Prepared by the Monitoring & Science Unit, Riverbend Catchment Authority. Data are provisional until the statutory return is lodged.
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